CBAClean:A Comprehensive System for Recommending Data Cleaning Solutions through Cost-Benefit Analysis in Data Quality Management

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***Abstract*—The scale of data analysis tasks have increased, highlighting the critical importance of data quality. Data quality assessment and repair have become pivotal in data preparation. Despite the availability of numerous algorithms for data cleaning, these often focus on optimizing efficiency and minimizing labor costs, neglecting the explicit relationship between data quality management costs and benefits. This omission can lead to the failure of promising data analysis solutions. To address this, we propose CBAClean, a comprehensive system that integrates cost-benefit analysis into data cleaning. CBAClean aims to assist users in quantifying the costs of data quality management and providing optimal data cleaning solutions tailored to their needs. Key features include task-centered multi-perspective data quality assessment, a comprehensive data quality repair operator library, fine-grained human role division for effective cost control, and recommendation of optimal data cleaning solutions based on cost-benefit calculations. By incorporating cost-benefit analysis, CBAClean enhances the practical application of data quality management on real-world data governance platforms.**

I. Introduction

In the rapidly evolving landscape of data science, the complexity of data analysis tasks continue to escalate, and highlights the importance of data quality. Recent years have witnessed a widespread acknowledgment of the direct impact of dataset quality on its application value [4]. Poor-quality data can lead to erroneous decisions and significant economic losses. Consequently, data quality assessment and repair have emerged as critical components in data preparation [6]. Drawing from extensive experiences and lessons learned in data quality management scenarios, we are aware that data quality improvement are costly endeavors, often requiring substantial human time and effort. Over the past few years, numerous algorithms (e.g., Holoclean [8], Baran [5], HAIPipe [1]) for data preparation and cleaning have been proposed. These efforts primarily focus on addressing the following issues: (i) **Advanced and domain-specific automated data cleaning tools**: Aiming to enhance cleaning efficiency and accuracy [7], [9], and (ii) **Constructing high-quality data preparation models with human-in-the-loop**: Incorporating human cognition and processing capabilities into data extraction, cleaning, and annotation processes through human-machine collaboration, while minimizing labor costs [1].

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Despite improving the efficiency of data quality management to some extent, these tools primarily concentrate on optimizing algorithms to boost data cleaning or labeling efficiency and minimize labor costs. They fail to explicitly elucidate the relationship between the costs of data quality management and the actual benefits derived. This omission is crucial, as the inadequate estimation of costs associated with data quality management can lead to the failure of many promising data analysis solutions. Therefore, current approaches to data quality management must not only consider the design of intelligent algorithms but also assess whether the costs invested in quality management translate into sufficient performance improvements in analysis results. Achieving this goal is non-trivial, primarily due to the inherent complexity of data quality management, the diversity of cost components, and the subjectivity and uncertainty in benefit assessment [2]. Specific challenges include:

- **Effective data quality evaluation for specific analysis tasks**: *How to assess the specific impact of data quality on analysis models?*
- **Scientific quantification of data quality management costs**: *How to scientifically and effectively quantify and account for the costs of data quality management, particularly labor costs?*
- **Constructing a cost-benefit assessment framework**: *How to determine the optimal cost investment strategy by comprehensively considering multiple factors to balance data quality management costs and quality repair benefits?*

Addressing these challenges, we propose a task-oriented data quality management cost and quality improvement benefit evaluation system CBAClean, building upon our previous extensive exploration of data cleaning [3], [6], [9]. CBAClean aims to assist users in quantifying the costs of data quality management and providing optimal data cleaning solutions that consider costs based on user needs. The key features of this system are summarized as follows:

(1) **Task-centered multi-perspective data quality assessment**: Ensuring the comprehensiveness and accuracy of assessments.

(2) **Comprehensive data quality enhancement operator library**: Meeting diverse data cleaning requirements.

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(3) **Fine-grained human role division and refined labor cost calculation**: Achieving effective cost control.

(4) **Recommendation of optimal data cleaning solutions**: Based on a comprehensive calculation of data quality management costs and improvement effects.

CBAClean not only effectively incorporates labor costs into the data quality management process but also provides additional insights for incremental data management services on actual data governance platforms through cost-based quality improvement calculations.

II. System Overview

Figure 1 outlines the architecture of CBAClean, primarily divided into the following core modules:

• *User Interaction*: This module enables users to upload datasets and input specific requirements related to quality assessment, various indicators for analysis models, and other relevant parameters.

• *Data Quality Assessment*: CBAClean has a multi-perspective data quality assessment model especially for analytical tasks. It provides a data quality dimension library, allowing users to evaluate data quality from multiple aspects, including completeness, consistency, accuracy, and currency, etc [9]. Additionally, a rule-driven error detection model is implemented to accurately identify fundamental data quality issues such as business rule violations, outliers, missing values, and spelling errors. More importantly, CBAClean further achieves task-specific quality assessment by integrating data characteristics, domain knowledge, and task requirements to evaluate the impact of different quality dimensions on the task and their sensitivity to quality issues. For example, CBAClean deeply analyzes whether the missing rate in a dataset significantly affects a prediction task. This tightly coupled quality assessment strategy with analytical tasks assists users in thoroughly analyzing whether the current data quality can support reliable computations for their tasks and in identifying existing data quality deficiencies.

• *Data Quality Improvement Cost Quantification*: This module is responsible for quantifying the cost required to enhance dataset quality. It first analyzes the quality assessment results to identify specific issues and their severity in data. Then, using predefined cost models, it calculates the resource investment needed to address these issues. In terms of data quality improvement operations, a comprehensive data quality improvement operator library is constructed, integrating various advanced cleaning operators from the current data cleansing domain, such as missing value imputation and data repair based on functional dependencies. Furthermore, an artificial operation library is maintained, predefining multiple methods for human involvement in data quality improvement, such as directly annotating detected quality issues, directly repairing detected erroneous data, and selecting candidate repair results generated by the operator library to solve problems that automated cleaning algorithms cannot address. In terms of cost quantification, CBAClean supports cost allocation for each atomic operation of data quality improvement. For automated algorithms, CBAClean provides cost calculation functions, such as quantifying cost in proportion to the number of operator invocations. For manual operations, more refined cost calculations [1], [2] are implemented based on different artificial roles (e.g., general labor, domain experts) and operation types to ensure the reasonableness of cost quantification.

• *Benefit Measurement and Cleaning Plan Generation*: This module analyzes the benefits of data quality improvement plans for analytical tasks through interactions with the quality assessment model and cost quantification model. Specifically, it calculates the improvement effects of analysis tasks on key indicators such as accuracy and recall after applying different cleaning plans. In addition, tailored to user requirements (e.g., expectations for expert user involvement and model improvement effects), the module constructs an optimization problem model and uses search algorithms to find combinations of multiple cleaning strategies, aiming to identify the cleaning plan with the lowest management cost and the highest benefits.

Users can view these recommended plans and select the most suitable one for execution based on their needs. CBAClean also supports a feedback mechanism for plan execution, allowing users to flexibly adjust the plan according to actual situations during execution. Once users provide feedback and adjustment suggestions, CBAClean re-initiates the cost quantification and benefit measurement processes, searches for and recommends optimal cleaning plans based on the udpated feedback results, thereby ensuring the flexibility and personalization of the data cleaning process.

III. Implementation Details

We introduce three core technologies of CBAClean, which encompass the generation of benchmark cleaning solutions based on quality assessment, human-centric cost quantification, and the calculation of optimal solutions under the cost-benefit model.

A. Generation of Benchmark Cleaning Solutions for Data Quality Management

Given a dataset $\mathcal{D}$ and a machine learning model $\mathcal{M}$ selected by the user from our algorithm library, the system initiates a comprehensive assessment of data quality. Let $\mathcal{D}$' represent the dataset after repair, where various quality issues have been identified. Based on user-defined parameters θ related to data quality and performance requirements, the system employs cleaning operators O_i from the cleaning operator library, where $i = 1, ..., n$ and n is the total number of operators. After applying these cleaning operations, the system computes the value function $U(S)$ of the dataset, which can be formally expressed as: $U(S) = f(\mathcal{D}', O_1, ..., O_n, \theta)$ where f is a function that quantifies the enhanced quality and utility of the dataset.

To assess the contribution of each cleaning step, the system calculates the Shapley value ϕ_i for each operator O_i: $\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n-|S|-1)!}{n!} [U(S \cup \{i\}) - U(S)]$ where N is the set of all operators and S is a subset of N. Additionally, the system computes the proportion of data altered, α_i, after

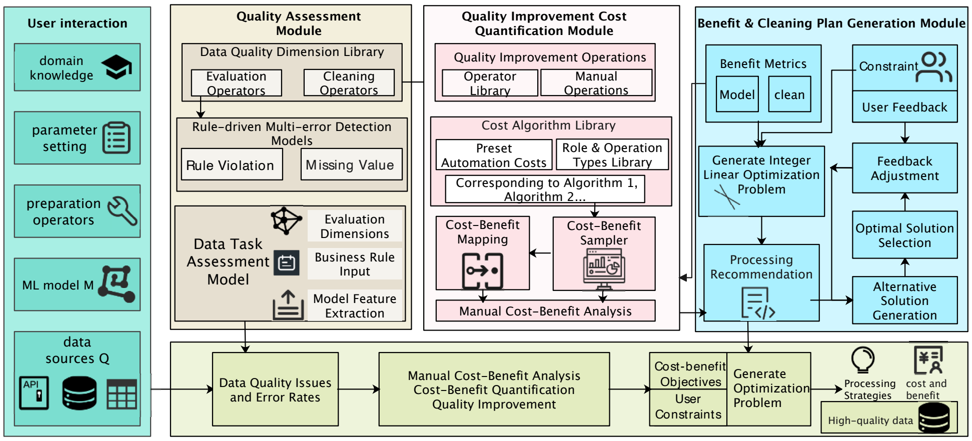

Fig. 1. CBAClean system overview.

executing each human-machine integrated step involving operator O_i: $\alpha_i = \frac{|\mathcal{D}'_i - \mathcal{D}'|}{|\mathcal{D}'|}$ where $\mathcal{D}'$ represents the dataset after the application of operator O_i.

Leveraging the user-preset threshold x for the extent of allowable human involvement, CBAClean identifies steps with $\phi_k(S) \cdot \alpha_k > x$ as candidates for recommended human participation. Based on these computations, CBAClean generates a benchmark data cleaning solution $\mathbf{P}$, which is formulated as: $\mathbf{P} = \{(O_k, \textit{Human Involvement?}) | \forall k, \phi_k(S) \cdot \alpha_k \leq x \Rightarrow \textit{No}, \text{otherwise} \Rightarrow \textit{Yes}\}$. This solution $\mathbf{P}$ serves as a starting point, balancing automation with human intervention to optimize the further cleaning process based on cost-benefit considerations.

B. Human-Centric Cost Calculation

Labor costs represent a crucial aspect of cost calculation within CBAClean. To address this, we have devised a sophisticated method for estimating human costs, focusing on both the roles involved and the time spent. In terms of time, CBAClean maintains a historical record of the unit time u_{ij} required for each type of data quality improvement operation j within task i. Based on the number of man-hours H_{ij} needed for these tasks (e.g., data annotation, error detection and repair), CBAClean computes the time cost $C_{\text{time},ij}$ for each operation with $C_{\text{time},ij} = u_{ij} \cdot H_{ij}$. With respect to human resources, the cost $C_{\text{labor},ij}$ is calculated considering the varying levels of expertise E_k among personnel k involved in step j of task i. This is given by: $C_{\text{labor},ij} = \sum_{k \in \text{personnel}} (E_k \cdot t_{ijk})$, where t_{ijk} represents the time spent by personnel k on step j of task i. The total cost $C_{\text{total},i}$ for task i is then computed as the sum of time and labor costs, considering concurrent tasks if applicable: $C_{\text{total},i} = \sum_j (C_{\text{time},ij} + C_{\text{labor},ij})$.

CBAClean further generates a model performance function S for each data cleaning solution, taking into account the varying levels of human involvement. By analyzing historical data and employing active learning techniques to extract features from the dataset, CBAClean is able to accurately assess the effectiveness of each solution. This process can be formalized as: $S(x) = \beta_0 + \beta_1 x_1 + ... + \beta_n x_n$, where $x_1, ..., x_n$ represent different factors affecting the model performance (e.g., degree of human involvement, type of data quality issues, etc.), and β_is are coefficients determined through regression analysis. For specific data processing steps, CBAClean simulates varying levels of human involvement on a sampled dataset and uses regression analysis to derive the model performance function S. This process enables the derivation of a cost-benefit correspondence.

C. Cost-Benefit Model for Optimal Solution Calculation

Upon receiving the baseline solution $\mathbf{P}_b$, CBAClean introduces cost-benefit optimization objectives and constraints (i.e., maximum cost and minimum benefit). To further refine the baseline solution within the feasible region, CBAClean enumerates a set of potential combined optimization solutions $Cl = \{P_i\}$. The performance Φ_{val} of each solution is predicted using a neural network, and the optimal solution $P_{\max}$ is selected as follows: $P_{\max} = \arg\max_{\mathbf{P}_i \in Cl} \Phi_{val}(\mathbf{P}_i)$.

To provide the most effective data cleaning solution, CBAClean specifically considers the following objectives:

Solution 1: Minimizing cost while maintaining accuracy. This solution aims to return a data preparation plan $\mathbf{P}$ that minimizes the overall human cost $cost(N_h, \mathcal{Q}, \mathbf{P})$, given a baseline plan $\mathbf{P}_b$. The plan must satisfy the conditions $acc(\mathcal{Q}, \mathcal{M}, \mathcal{P}) \geq acc(\mathcal{Q}, \mathcal{M}, \mathcal{P}_b)$ and $cost(N_h, \mathcal{Q}, \mathcal{P}) \leq cost(N_h, \mathcal{Q}, \mathcal{P}_b)$, where acc represents accuracy, $\mathcal{Q}$ is the data quality metric, $\mathcal{M}$ is the model, and N_h denotes the human resources involved.

Solution 2: Maximizing accuracy while maintaining cost. This solution seeks to return a data preparation plan $\mathbf{P}$ that maximizes the final model performance $acc(\mathcal{Q}, \mathcal{M}, \mathcal{P})$, given a baseline plan $\mathbf{P}_b$. The plan must satisfy the conditions $acc(\mathcal{Q}, \mathcal{M}, \mathcal{P}) \geq acc(\mathcal{Q}, \mathcal{M}, \mathcal{P}_b)$ and $cost(N_h, \mathcal{Q}, \mathcal{P}) \leq cost(N_h, \mathcal{Q}, \mathcal{P}_b)$.

Solution 3: Minimizing human cost in specific stages. In many cases, while the overall human cost $cost(N_h, \mathcal{Q}, \mathcal{P})$ may be low, the human cost in a particular stage can be disproportionately high. For instance, while the overall human cost might account for 10% of the total, data annotation might consume 70% of this cost. In such scenarios, users often desire to minimize human costs across all stages while

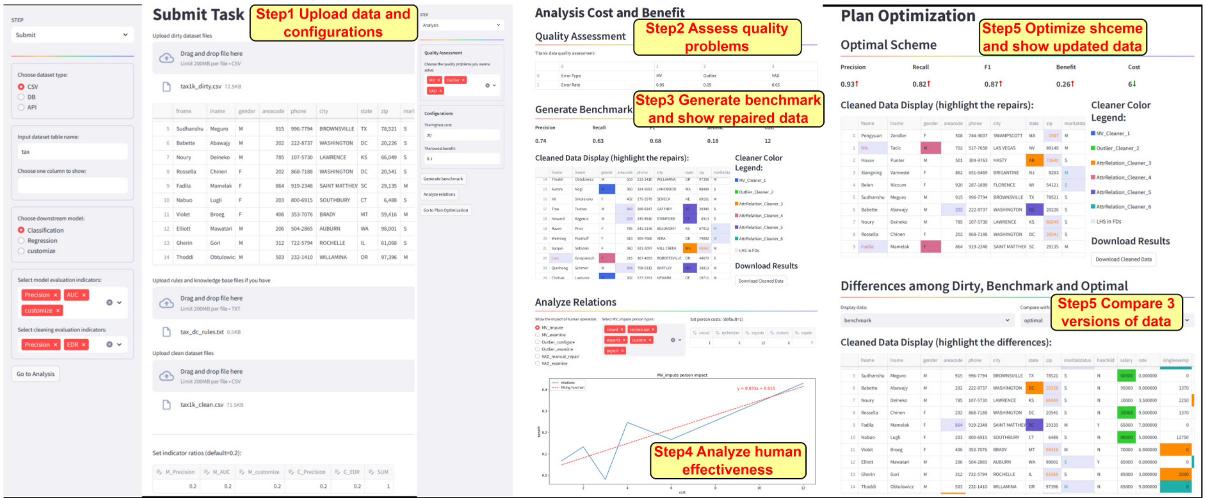

Fig. 2. Pages demonstration in CBAClean.

ensuring that the overall cost remains at most equal to and the overall accuracy is at least as high as the baseline. Once the optimal solution is identified, it is implemented in the actual data preparation process. The system continuously tracks the effectiveness and cost, adjusting the solution based on user feedback for iterative optimization. This ensures that the model performance is maximized without exceeding the expected cost. Ultimately, CBAClean outputs the data processing plan, the associated costs and benefits, and the high-quality data obtained through the processing plan.

IV. Demonstration

We intend to demonstrate all the aforementioned steps for a comprehensive data cleaning solution recommendation workflow, as shown in Fig. 2.

▷ **Dataset and cleaner library configuration**. Users can upload data through CSV files, database, or API interfaces. Additionally, users are required to upload knowledge bases or rule sets, which serve as the foundation for subsequent data preparation and analysis. Depending on the downstream model type and the availability of clean datasets, users can freely choose model evaluation metrics and data cleaning metrics. For each selected metric, users can assign specific weights to reflect its importance in the data cleaning process.

▷ **Data quality issue detection**. Upon submitting a task, the system enters the analysis phase. In this step, users select the quality issues they wish to detect and resolve, such as missing values, outliers, and duplicate records. The system automatically detects these quality issues within the dataset based on user selections.

▷ **Benchmark solution generation**. For the detected quality issues, CBAClean generates a benchmark data cleaning solution. This solution includes a series of data processing steps aimed at addressing the selected quality issues. After processing, the system displays the dataset processed by the baseline solution for user review.

▷ **Cost-benefit analysis and model performance prediction**. In this step, CBAClean simulates varying degrees of human involvement in processing the sampled dataset, based on the different types of personnel (e.g., general staff, domain experts) selected by the users. Through regression analysis, CBAClean derives a model performance function for labor costs in each stage, which predicts changes in model performance under different levels of human involvement.

▷ **Cleaning solution optimization and display**. Users can choose to download the dataset processed by the baseline solution or proceed with solution optimization. CBAClean employs advanced algorithms to optimize the baseline solution, further enhancing data quality and model performance. After optimization, CBAClean displays the improvements in metrics and the repaired dataset.

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